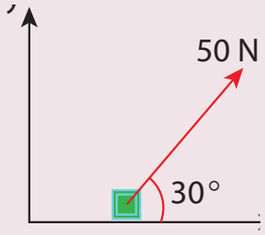
IV. Numerical Problems

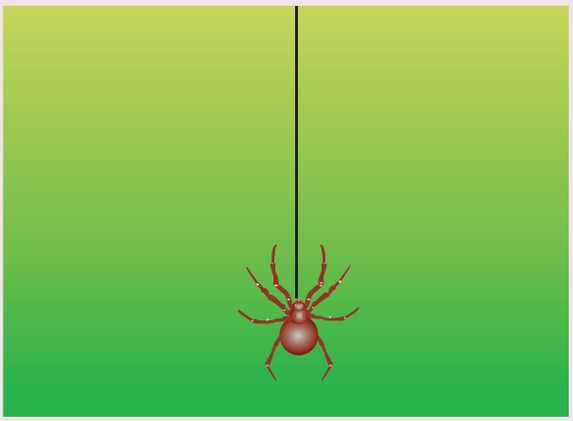
1. A force of 50N act on the object of mass 20 kg. shown in the figure. Calculate the acceleration of the object in x and y directions.



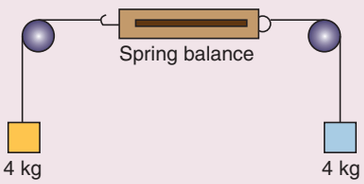
Ans: a_x=2.165 ms^{-2}; a_y=1.25 ms^{-2}

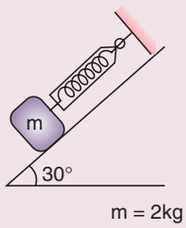
2. A spider of mass 50 g is hanging on a string of a cob web as shown in the figure. What is the tension in the string?

Ans: T=0.49N



3. What is the reading shown in spring balance?





Ans: Zero, 9.8 N

4. The physics books are stacked on each other in the sequence: +1 volumes 1 and 2; +2 volumes 1 and 2 on a table.

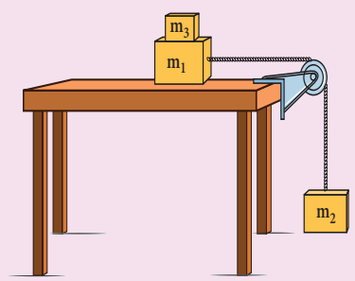
a) Identify the forces acting on each book and draw the free body diagram.

b) Identify the forces exerted by each book on the other.

5. A bob attached to the string oscillates back and forth. Resolve the forces acting on the bob in to components. What is the acceleration experience by the bob at an angle θ.

Ans: Tangential acceleration = g sinθ ; centripetal acceleration = $\frac{(T-mg\,cos\theta)}{m}$.

6. Two masses m_1 and m_2 are connected with a string passing over a frictionless pulley fixed at the corner of the table as shown in the figure. The coefficient of static friction of mass m_1 with the table is μ_s. Calculate the minimum mass m_3 that may be placed on m1to prevent it from sliding. Check if m_1=15 kg, m_2=10 kg, m_3=25 kg and μ_s =0.2



Ans: $m_3 = \frac{m_2}{\mu_s} - m_1$, the combined masses m_1+m_3 will slide.

7. Calculate the acceleration of the bicycle of mass 25 kg as shown in Figures 1 and 2.

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